Opening Questions

Sports

- Who is the best quarterback of all time?
- Are baseball pitchers throwing harder than ever?
- How much more do male soccer players earn than females?
- How common is it for former Olympic athletes to become coaches?
- How much does an extra inch of height help a basketball player?

Pop Culture

- What percentage of people have seen the movie that won last year's Best Picture Award?
- Who tends to be more popular: bands or solo singers?
- Are younger actors paid more than older actors?
- Are movies with female leads as profitable as movies with male leads?
- Does winning a Grammy increase sales?

Politics

- Is "Stop and Frisk" a racist policy?
- Do Republican politicians tend to come from different states than Democratic ones?
- Do people in countries that have universal healthcare live longer than people in countries that don't?
- Was press coverage slanted for or against a particular candidate?

Education

- Do small schools perform better than large ones?
- Which has a stronger correlation with student achievement: race or wealth?
- Do bilingual classes result in better outcomes for ESL/ELL students?

What Questions Can You Answer with the Given Data?

The following is a dataset of a bicycle rider's training rides.

date	miles	time (w/stops)	weather	average speed	max speed
04/10/2018	10	44	"cloudy"	13	30
05/30/2018	15	66	"sunny"	13.5	22
06/12/2018	12	61	"rainy"	11.2	25
07/04/2018	24	103	"sunny"	14	26
07/12/2018	24	120	"windy"	12.5	26

1) Decide whether each questions below *can* or *cannot* be answered with the given data and circle your selection.

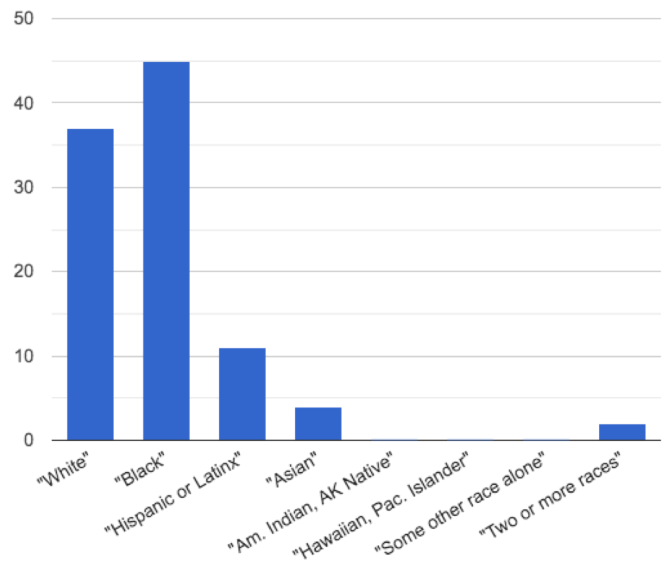
2) In the space provided below, explain *how* you could answer the question using the data or *why you cannot* answer the question.

Question	Answered by this dataset?
How many miles did the cyclist ride June 12th?	*Yes _No*
What tire pressure produces the highest average speed?	*Yes* _No
What is the average time it takes this cyclist to ride 1 mi?	*Yes _No*
Does this cyclist ride slower when it is rainy?	*Yes* _No
Does this cyclist ride faster when they are late to an appointment?	*Yes* _No
How many miles has the cyclist ridden in total as part of their training?	*Yes _No*

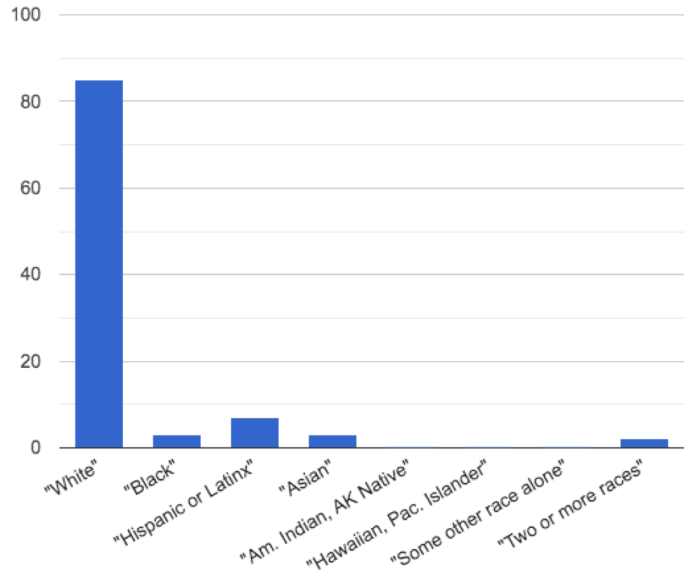
Are there any questions that you could find the answers to more than one way?

Bar Chart - Notice and Wonder

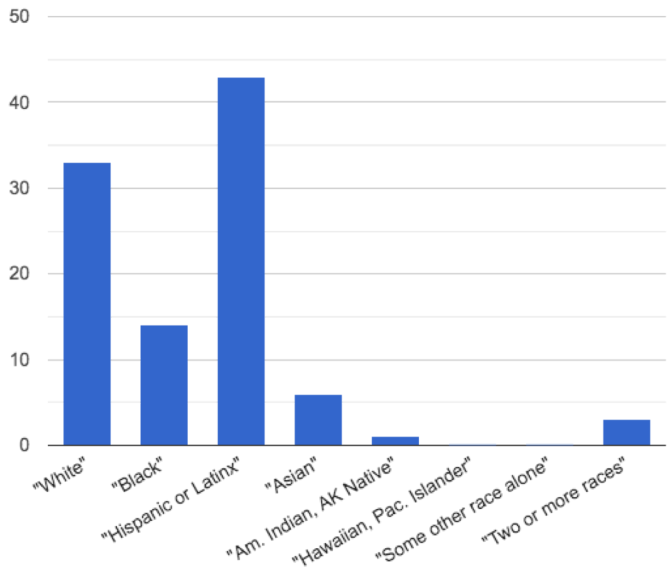
Washington, D.C. Public Schools



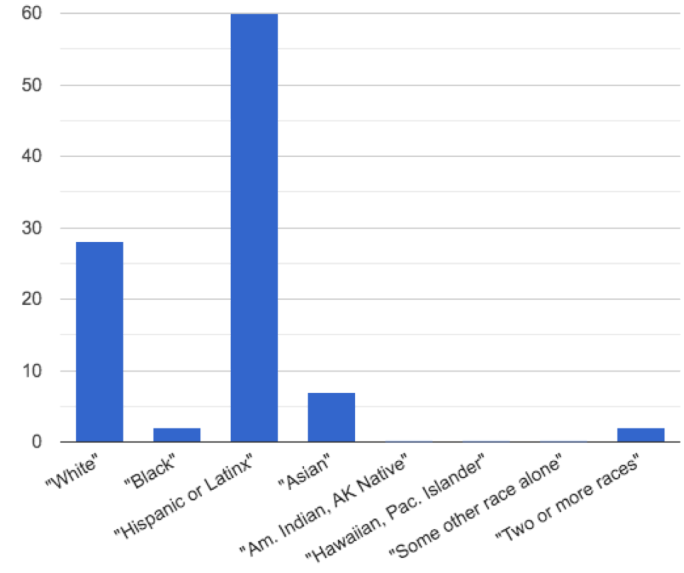
West Warwick Schools, RI



Providence Schools, RI



Oxnard Union HS District, CA



What do you Notice?	What do you Wonder?

Question Types: Animals

A subset of the whole Animals Dataset is shown in the table below.

name	species	sex	age	fixed	legs	pounds	weeks
Sasha	cat	female	1	false	4	6.5	3
Sunflower	cat	female	5	true	4	8.1	6
Felix	cat	male	16	true	4	9.2	5
Sheba	cat	female	7	true	4	8.4	6
Billie	snail	hermaphrodite	0.5	false	0	0.1	3
Snowcone	cat	female	2	true	4	6.5	5
Wade	cat	male	1	false	4	3.2	1
Hercules	cat	male	3	false	4	13.4	2
Toggle	dog	female	3	true	4	48	1

Using this table - or the full dataset - write three questions of each type below.

	Question	Type
1		Lookup
2		Lookup
3		Lookup
4		Arithmetic
5		Arithmetic
6		Arithmetic
7		Statistical
8		Statistical
9		Statistical

Expression Cards

Print and cut these out, for use with the `Filter` Transformer activity in this lesson. You can make up your own using the blanks provided at the bottom!

Do they wear glasses? `Student -> Boolean` consumes a student from the dataset, produces true if they are wearing glasses	**Are they taller than you?** `Student -> Boolean` consumes a student from the dataset, produces true if they are taller than you
Do they have long hair? `Student -> Boolean` consumes a student from the dataset, and produces true if they have long hair	**Do they have red hair?** `Student -> Boolean` consumes a student from the dataset, and produces true if they have red hair
Are they shorter than you? `Student -> Boolean` consumes a student from the dataset, and produces true if they are shorter than you	**Are they smiling?** `Student -> Boolean` consumes a student from the dataset, and produces true if they are smiling
Does their name begin with a vowel? `Student -> Boolean` consumes a student from the dataset, and produces true if the first letter of their name is a vowel	**Are they wearing jeans?** `Student -> Boolean` consumes a student from the dataset, and produces true if they are wearing jeans

What Table Do We Get?

The table below represents four animals from the shelter:

name	sex	age	fixed	species	pounds
`"Toggle"`	`"female"`	`12`	`true`	`"dog"`	`48`
`"Fritz"`	`"male"`	`4`	`false`	`"dog"`	`92`
`"Nori"`	`"female"`	`6`	`true`	`"dog"`	`35.3`
`"Sunflower"`	`"female"`	`2`	`false`	`"cat"`	`51.6`

Below is a list of saved Transformers. What table do we get, after applying each one? *Match* each expressionto the table we get .

`order-by-age`
Transformer: Sort
Attribute to Sort By: age
Direction: ascending

1

`order-by-weight`
Transformer: Sort
Attribute to Sort By: pounds
Direction: descending

2

`filter-is-old`
Transformer: Filter
Formula to Filter By: Age > 10

3

`filter-is-fixed`
Transformer: Filter
Formula to Filter By: `Fixed = true`

4

`filter-is-dog`
Transformer: Filter
Formula to Filter By: `species = "dog"`

5

`build-is-dog`
Transformer: Build Attribute
Formula for New Attribute Values: `species = "dog"`

6

`build-is-fixed`
Transformer: Filter
Formula to Filter By: `fixed = true`

7

A Produces a table with Toggle, Fritz, and Nori - but not Sunflower.

B Produces a table of all four animals, sorted youngest-to-oldest

C Produces a table, with only Toggle.

D Produces an identical table with an extra column called "dog", whose values are true, true, true, false

E Produces a table containing only Nori and Toggle.

F Produces a table with all four animals, sorted from heaviest to lightest.

G Won't run: will produce an error

Table Transformations with Chaining

You have created and saved the following Transformers. (Read them *carefully!*)

`filter-is-fixed`	`filter-is-young`	`filter-is-female`	`build-kilos`
Transformer: Filter Expression: Fixed = true	Transformer: Filter Expression: Age < 4	Transformer: Filter Expression: Sex = “female”	Transformer: Build Attribute Name of new attribute: Kilos Expression: Weight (lbs) / 2.2
`filter-is-heavy`	`filter-is-dog`	`filter-is-cat`	
Transformer: Filter Expression: Kilos > 25	Transformer: Filter Expression: Species = “dog”	Transformer: Filter Expression: Species = "cat"	

Solve the word problems below using the Transformers listed above.

	Prompt	Does order matter?	List the transformers in order.	Use function notation to predict the table’s name.
1	Create a table of all female cats.	No	Apply `filter-is-female` . Apply `filter-is-cat` .	filter-is-cat(filter-is-female(animals-table))
2	Create a table of all young, fixed animals.			
3	Create a table of all female dogs, converting their weight to kilograms.			
4	Create a table of all heavy dogs.			
5	Create a table of all young, fixed dogs, sorted from lightest to heaviest.			

Predictions from Samples

1) Use the Sampler plugin to create the following tables and give them the specified names. (Re-open Sampler *before* creating each table to avoid getting one massive hierarchical table!)

- 10 animals randomly selected without replacement ⇒ tiny-sample
- 20 animals randomly selected without replacement ⇒ small-sample
- 40 animals randomly selected without replacement ⇒ medium-sample
- 80 animals randomly selected without replacement ⇒ large-sample

2) Make a bar chart of the species in the tiny sample.

- What animals are in the sample? _____
- Create a new random tiny-sample. Create another bar-chart of species in the tiny-sample. What animals are in the sample? _____
- Create another random sample and make another bar-chart of species in the tiny-sample. Based on these samples, how many species of animals do you think are at the shelter? _____
- Which species do you think there are the most of at the shelter? _____

3) What did you learn from taking multiple samples that you wouldn't have known if you'd only taken a single sample?

_____

4) Now use the small sample to make a bar chart of the species.

- What animals are in the sample? _____
- Create a new random sample and make another bar chart of species in the small sample. What animals are in the sample? _____

5) Now that you've seen the small sample, how has your sense of the distribution of the species changed?

_____

6) Now use the medium sample to make a bar chart of the species. If there are about 400 animals at the shelter, how many of each species would you predict there to be?

_____

7) Now use the large to make a bar chart of the species. If there's anything you'd like to change about your prediction now that you've seen the large sample, record it here.

_____

8) Let's see how accurate your prediction is… When you're ready, make a bar chart of Animals Table 2.

- Which predictions were closest? _____
- Which predictions were off? _____
- Were there any surprises? _____

9) In the real world, we usually don't have access to a whole dataset to check predictions against! How could we test…

- *Every giraffe on the planet?*
- *Everyone who has ever come in contact with a covid-positive person?*
- *Every person who identifies as queer?*

What strategies can we use to make sure that predictions from samples are as close to accurate as possible?

_____

_____